

# Evaluating LLM-based Process Model Generation: From Academic Prototypes to Production-Scale Software Capability

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## Abstract

The advent of generative artificial intelligence and, more specifically, large language models (LLMs), has led to a plethora of applied research in software and information systems engineering. In the sub-field of business process management, a problem that is frequently explored in this context is the automated generation of graphical process models from text, also known as *text-to-model*. In this paper, we give a comprehensive overview of five existing LLM-based text-to-model approaches, including two novel methods that have been developed as part of successful industry-scale product development efforts. We provide a comprehensive comparison of the approaches, given a fixed LLM and five ground-truth datasets comprising over 300 description text-model pairs. We also discuss how academic approaches interact with—and sometimes differ from—industry efforts that implement LLM-based text-to-model approaches in practice. Here, the full productization of a text-to-model capability as part of SAP standard software serves as our case study. Our findings show that [T: [TODO, what they show and why it matters.](#)]

**Keywords.** Business process management, conversational process modeling, process modeling software, generative artificial intelligence

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# 1 Introduction

Process models describe how work is—or should be—executed in order to accomplish organizational objectives. Graphical process models in languages such as the open standard-based Business Process Model and Notation (BPMN) [OMG13] are crucial artifacts for business process management (BPM). For example, they can help identify shortcomings in current process executions or desirable changes in operations that are expected to lead to better performance. Even data-driven BPM techniques such as process mining often rely on process models as input artifacts, for instance for conformance checking, or generate process models as output (of *process discovery* algorithms) [T: CITE]. However, creating process models is effortful [T: CITE]: data for auto-generating process models is often not readily available at the right granularity level, and manual process model creation requires process modeling skills as well as substantial domain expertise. To reduce the manual effort required for process model generations, software-based approaches that generate process models from text, sometimes called *text-to-process* approaches, have been developed over the past decades. Initially, these approaches were rule-based [T: CITE], and thus had severe limitations in their ability to deal with the nuance and ambiguity of natural language. With the advent of large language models (LLMs), research on text-to-process has resurged. LLM-based text-to-process approaches have been developed and successfully been deployed in process management products. While academic text-to-process approaches are often presented alongside empirical evaluations of their performance, the applied evaluation approaches tend to differ across approaches, notably due to differences in the applied datasets, as well as in the similarity measures that are used to compare generated and ground-truth models. Accordingly, it is not clear how the devised approaches differ, both conceptually and empirically (in terms of performance).

The objective of this paper is to shed more light on the differences between approaches. This is achieved on the *conceptual* level by providing an overview of [T: how many?] existing approaches in the literature, highlighting fundamental similarities and differences; the collection of approaches is extended by [T: how many?] additional approaches that have been developed in an industry productization setting. On the *empirical* level, a comprehensive comparative performance evaluation is conducted, utilizing several ground-truth datasets and similarity notions/measures. The evaluation allows for a nuanced comparison that goes beyond arguably simplistic 1-dimensional precision and recall [T: we can consider rephrasing]. Finally, we discuss the results in light of real-world product decisions and trade-offs made in the course of the deployment of text-to-process capabilities as part of the generally available product offering of a large enterprise software vendor.

[T: Not sure if we need “the rest of the paper is organized as follows”.]

1. Description of Problem Domain
2. (a) Entry barrier for process modeling in business context
  - (b) Vast amount of knowledge documented in textual form or knowledge in the organization/domain experts without BPM background. A survey conducted in 2023 at a SAP Signavio event with 35 BPM tool usage participants from small (5+ employees) to large enterprises (100.000+ employees) (multiple participants per company possible) resulted in Text2Process extraction as the most exciting AI Capability (45% of votes compared to Image2Process with 27%, Extraction of Process Dictionary Elements with 16% and Process model Summary generation (10%). 37% of participants marked Text2Model as a "game changer", 48% as "important" and 14% as "nice to have". [C: Either here or in the outlook, we could say that 50% reported to have more than 500 documents, mostly word, pdf and powerpoint containing process relevant information.]
  - (c) Ambiguity in process modelling
  - (d) (further aspects to be drawn from existing papers)

## 2 Industry Requirements

In recent years, the rise of LLMs has emerged transformations across multiple domains. Business process management (BPM) has likewise been influenced by this trend and is beginning to explore LLM capabilities across all stages of the BPM lifecycle [HWR25].

One area that has gained particular attention due to LLMs' ability to understand and generate natural language is the transformation of unstructured text into structured process models.

@christian, @timmy: add info about survey and participants (industry background, positions, etc.): performed in SAP in 2023 as a part of a Berlin forum with 35 participants ...

According to the survey, one of the most exciting AI-capability related to process modeling was text-to-process transformation (45%), ranking ahead of other options such as extracting dictionary-relevant entities (e.g., it-systems) (16%), process model extraction (27%), and suggesting a short summary of a process model (10%). A majority of participants (40%) also rated this feature as "important", 28% even describing it as "game changing", with only 28% considered it as "nice-to-have".

This trend can also be observed in the academic field. According to a survey on LLM applications in BPM (Mapping the Landscape: Exploring LLM Applications in BPM), over 50% of the reviewed papers focused on process model discovery, particularly on the generation of business process models [ETdROR24].

LLM-based approaches to process model generation help overcome challenges that hindered earlier methods for text-to-model transformation, such

as (a) handling unstructured text, (b) grammar issues and variations in writing style, (c) domain-specific terminology, and (d) the inability to modify the model after it has been generated [AOL24]. Despite the fact that LLMs reduce complexity of traditional NLP-based text-to-model pipelines, enabling more vivid, human-like communication between the user and the system, LLM integration brings its own challenges and limitations that have to be taken into account [AMH25].

Problems like hallucinations, misalignment between LLM behavior and human intention, measuring the quality of the generated models [WSBM25], along with reliability and accuracy, performance and hardware requirements, data privacy and security, integration with existing systems and processes, ethical considerations, biases, transparency, and explainability [AMH25] introduce new issues that developers and users have to be careful about and address.

However, not all of these challenges are equally relevant for research prototypes and final products intended for use in industry. For instance, academic research often focuses on achieving high accuracy, typically measured by comparing generated artifacts to a ground truth. In contrast, this approach is less applicable to final products, since in real-world scenarios, a definitive ground truth is often unavailable. While accuracy is often valued, it is not always the primary objective of these applications. According to the survey mentioned above, most of the participants answered that by utilizing text-to-process functionality, they would expect a rough draft of a business process (37%), a summary of relevant process information (e.g., a list of activities) (33%), and only then a process fully compliant with BPMN (27%) (with the remaining 2% choosing other options).

Experimental and measurement bias can also be an issue, since in the research context evaluation is more a snapshot in time, whereas in productization, consistently stable and reliable performance is of major importance. This also leads to differences in testing and evaluation techniques, since the purpose of a prototype is to test the idea and validate user experience, while the aim of the final product is to provide a consistently satisfying user experience (aspects that are often considered secondary in academic prototypes). Additional robustness analysis might be needed. Furthermore, industry users may be more interested in integrating generated models with existing BPM tools (e.g., generating models in BPMN 2.0 format rather than in some intermediate representation, which is often the case in research prototypes).

These contrasts result in a situation where the effectiveness and advanced capabilities of LLMs are demonstrated in ongoing research, but are not yet suitable for productization. In addition, highly promising results in lab-only settings with simple, small-scale, and already well-familiar datasets may not be translatable to the real world scenarios [LTLL24]. Thus, it is crucial to evaluate existing solutions through the prism of both perspectives to set appropriate expectations.

### 3 Methodology

The methodology used for the comparative evaluation presented in this paper comprises four steps: (1) data collection, (2) selection of relevant approaches, (3) artefact generation, and (4) evaluation and interpretation of results. During data collection, relevant datasets containing both process descriptions and associated process models are gathered. Then, based on a review of the existing literature, a list of relevant and available approaches is compiled. Next, we apply the selected approaches to the collected datasets to generate new artefacts, i.e., process models derived from the provided process descriptions. Finally, we evaluate the performance of the selected approaches using predefined measures, capturing, on one hand, process model complexity, and on the other hand, the similarity of the generated process models to both the source process descriptions and the associated ground-truth process models.

In this section, we provide a detailed overview of the collected datasets, selected approaches and employed LLMs, as well as a list of the utilized measures.

#### 3.1 Datasets

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For the evaluation we utilize 5 different datasets. Every dataset consists of process descriptions and associated process models, with all descriptions and models provided in English. Process descriptions serve as input for the further generation of models, and process models are utilized for evaluation as a ground-truth reference. For the basic overview of the dataset characteristics see table 3.1.

**Domains.** The dataset comprises 25 examples spanning five distinct domains: GDPR, healthcare, logistics, manufacturing, and tourism. Each example includes process models utilizing core BPMN elements such as start and end events, tasks, parallel and exclusive gateways. Process descriptions are created from these models through a template-based method utilizing the PromptGen tool<sup>1</sup>, as detailed in [KMKRM24].

**MAD150.** The dataset consists of 150 examples extracted from the MAD dataset [LNL<sup>+</sup>23], i.e., 10 examples for each business category. The dataset contains simple process models with up to 20 tasks. The models include fundamental BPMN elements, such as start and end events, tasks, parallel gateways, and inclusive gateways. The process descriptions accompanying the models are generated using a template-based method, as detailed in [LNL<sup>+</sup>23].

**PET35.** The dataset includes 35 examples sourced from the PET dataset [BvdAD<sup>+</sup>22]. Certain existing examples from the original dataset were excluded from the experiment due to inconsistencies and gaps that could result in potential misinterpretations or deviations in the process models. The process descriptions are provided in the artificial dataset. However, process models are derived manually from annotations of process descriptions provided in the artificial data set. These models incorporate BPMN elements such as start and end events, tasks,

<sup>1</sup><https://github.com/etm/promptgen>, last access: 2025-07-31

parallel and exclusive gateways, as well as pools and lanes.

**SAPSAM69.** The dataset consists of 69 examples sourced from the SAPSAM collection (see [SWS<sup>+</sup>22] for more details). Since the initial dataset contains process models in different modelling and natural languages, it was filtered based on quality criteria: BPMN 2.0 format, English language, valid syntax, a minimum of four activities, at least one lane, and descriptive labels ( $\geq 5$  characters). The process models include not only basic BPMN elements, but also pools and lanes and special events (e.g., timer, message, etc.). The initial dataset contains only process models. Thus, the process descriptions utilized here are generated from these models using LLM (gpt-4o) and subsequently verified by human reviewers to ensure their accuracy and quality.

**RealSet.** The dataset comprises 23 examples from six different domains (see [MK23] for more details). The process models are created and evaluated by process modelers based on textual process descriptions. Since the models were created manually, they represent one possible interpretation of the textual descriptions and may include slight deviations or additions. These variations reflect the modelers’ common sense, personal experience, and domain knowledge. The process models include not only basic BPMN elements, but also pools and lanes and special events (e.g., timer, message, etc.).

Table 1: Basic Dataset Information. Values are represented as averages per dataset. PD – Process Descriptions; S – Sentences; W – Words; Events – Start, End events; Gateways – Exclusive, Inclusive, and Parallel; Special Events – Timer, Message, Escalation, Conditional, Error, Terminate, and Cancel.

| Dataset  | # PD | # S  | # W   | Supported BPMN Elements                               | Consistency  |
|----------|------|------|-------|-------------------------------------------------------|--------------|
| Domains  | 25   | 19.6 | 275.4 | Tasks, Events, Gateways                               | high         |
| MAD150   | 150  | 8.4  | 136.2 | Tasks, Events, Gateways                               |              |
| PET35    | 35   | 8.9  | 140.8 | Tasks, Events, Gateways, Pools, Lanes                 | intermediate |
| SAPSAM69 | 69   | 10.8 | 148.0 | Tasks, Events, Gateways, Pools, Lanes, Special Events |              |
| RealSet  | 23   | 9.5  | 173.9 | Tasks, Events, Gateways, Pools, Lanes, Special Events | low          |

Since the datasets differ in their nature (i.e., the way how the data was created), they are categorized into three groups based on the level of consistency between process descriptions and their associated process models ranging from highly consistent model-text pairs to those with low consistency (see Table 3.1). This categorization allows for evaluating the real state of things by distinguishing between simple artificial examples and those that represent real-world scenarios. In addition, it helps to find the limitations of utilized methods and to obtain a full picture of their performance.

- **High.** In these datasets (Domains and MAD150) process descriptions were created using template-based method based on existing process models. This method ensures full consistency as well as full semantic alignments between the existing process model and the created process description. The quality of these descriptions in terms of readability and

human-friendliness is quite low. However such texts are very easy to parse and are easy to understand by LLM [KMKRM24]. Evaluation of such semi-structured data with high clarity might provide an intuition about performance of model generation process reducing or even isolating the impact of writing style.

- **Intermediate.** These datasets (PET35 and SAPSAM69) are widely used in the community. LLMs are most likely already familiar with them. They are created to represent an ideal scenario where process models and texts are highly consistent with each other and the process descriptions are written in a fluent language familiar to a non-modelling expert (something that is unlikely in real-world scenarios). The analysis of this data ensures the correctness of model generation approaches (i.e., proof of concept).
- **Complicated.** This data (RealSet) is highly inconsistent. Since process models were created based on provided process description by modeling experts, they are mostly more complex and might contain inconsistencies, ambiguities, and diverse interpretations. Analyzing this kind of data allows us to better capture the true state of the art, reflecting the real-world complexity of data, allowing better understanding of the LLMs limitations and utilized approaches, helping to make existing approaches more suitable for practical applications.

## 3.2 Approaches

Here, we provide an overview of N approaches selected for evaluation. These approaches were chosen because they all focus on generating process models from natural text, have been formally presented (e.g., as demos or conference papers), and are available as open-source tools. For each approach, we provide a brief description, specify the LLM used for generation, and list any existing limitations.

### 3.2.1 PromoAI

PromoAI [KBSvdA24] is a prompt-engineering based approach for the text-to-process task. It works by using GPT to convert a textual description into runnable Python code, which generates an object in the Partially Ordered Workflow Language (POWL) [KvZ23]. POWL allows to generate an intermediate representation of the model, which is structurally more simple than e.g. directly generating a graph. For prompt-engineering, they use advanced techniques such as role prompting and few-shot learning. In addition, the LLM is allowed to refine it's solution by iteratively generating a new solution, whenever the generated Python program has a syntactic or semantic error. After generating the POWL object, it can directly be converted into a BPMN model or a Petri net. It is worth noting, that this approach only generates models with sequence flows, gateways, and start or end events. Swimlanes or other more complex stencils are not generated.

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which model did we use?

### 3.2.2 Efficient LLM-Based Conversational Process Modeling (KLU)

Jan

[KS25]

### 3.2.3 AutoBPMN.AI (AutoBPMN)

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[C: Is there a paper for this approach/code that we can run? otherwise as agreed we would move this into related work right? ]

[N: Webpage: <https://autobpmn.ai/>. Paper: <https://ceur-ws.org/Vol-4032/>. The BPM Workshop Proceedings 2025 will be published as post-proceedings at Springer. And yes, I have the code. We have also implemented a rest API that can be used to generate models, that I'm using.]

### 3.2.4 A Decomposed Hybrid Approach to Business Process Modeling with LLMs

[N: This approach has to be listed only in the related work. In this section we describe only approaches that were used for the evaluation] [NEAA<sup>+</sup>25]  
Note: While this approach constitutes an interesting take on AI Assisted Process Modelling, we were not able to test it on the broad example datasets given that its embedded in a commercial tool without an open source repository.

potentially  
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work

### 3.2.5 SplitMiner Approach to Process Generation (SplitMiner)

This novel approach uses a simple representation of processes as set of potential sequences through the process. The task of the LLM is to generate sequences that represent the potential paths through a process. For an XOR-split, an additional sequence is created. For an AND gateway, all potential combinations are created.

Layouting.

Note: This approach is mainly introduced and discussed to outline the approach. Limitations were quite early discovered and not pursued in a commercial feature any further.

### 3.2.6 Annotated Process Trees (Trees)

This novel approach leveraging annotated process trees uses a recursive intermediate structure for the LLM to generate.

[C: @natalia - do we also cite Friedrich et al for completion even if we don't compare them.] [N: I would rather cite him only in related work, not to confuse people, who will read the paper. Here I would only list the approaches that are used in the evaluation. ]



### 3.3 Measures

To evaluate how well each of the approaches is doing, we have to consider multiple aspects that are related to (i) the basic features of the approach, (ii) the quality of the generated process models (process model-related measures), as well as to (iii) aspects that are related to the approach itself, like its computational performance and costs (performance-related measures).

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#### 3.3.1 BPMN Stencil Coverage - Basic Capability Dimensions

T2P v2 covers over 85% of BPMN elements.

These features serve to cover the basic industry requirements necessary for a BPMN tool to function effectively in enterprise environment and are more concentrated on usability, interoperability, scalability, and user experience. And thus, reflect the maturity of the approach for its practical usage. The selected capabilities are listed below, with a detailed overview presented in the Table 2.

- *Direct BPMN 2.0 Support*: Does the approach natively support BPMN 2.0 or use a custom/intermediate notation? (yes \no)
- *Stencils Covered*: How many BPMN elements are supported (e.g., tasks, events, gateways, pools)?
- *User Interface*: Is there a GUI provided? (yes \no)
- *Model Redesign via GUI*: Can users modify the generated model directly within the tool manually? (yes \no)
- *Model Redesign via Chat*: Can the user iteratively refine/adjust the model through natural language interaction? (yes \no)
- *Deployment Type*: Is it a standalone research prototype, demo, or integrated in a commercial platform? (standalone,demo,integrated)
- *REST API*: Does it offer a REST API for integration into other systems? (yes \no)
- *Input Format*: What type of input does it accept? (text, files, images, speech)
- *Output Format*: What type of output does it produce? (.bpmn, custom formats, images, text (i.e., text description of the generated model: model-to-text)).

c: Include measure to describe what subset of BPMN elements are covered. Include coverage of BPMN elements. TODO Christian - find out which workspace/-dataset they were calculated on?

fill the table

#### 3.3.2 Process Model-related Measures

Since we consider the generated process model as a graphical artifact, we focus on measurements capturing static properties of business processes [SGRV10]. First, we consider how complicated the generated model is (i.e., model complexity), dealing only with activity and control-flow complexity [Car05]. This means that we do not consider *data flow* and *resource* elements during evaluation, as existing approaches for process model generation using genAI are not able to capture these aspects of the process model. Process model complexity can provide insights on how difficult it is to analyze, understand or explain a process model [Car05] and also be used as a determinant of error probability of a business process [CMNR06].

Also, we consider *process model similarity*, to find out how complete the generated model is in comparison with the original text and the ground truth process model. Here, it is important to mention, that results are highly dependable not only on the effectiveness of the generation approach itself, but also on the quality of the available ground truth.

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Finally, since we our target output are process models in a well-established modeling standard (i.e., BPMN 2.0), we evaluate selected approaches based on symbolic rules for good process modeling, so-called *modeling conventions* to ensure clarity, sustainability and executability of generated process models<sup>2</sup>.

**Process Model Complexity.** According to a review provided by [Mel12] there is no singular definition of process model complexity; notably, the term is often used interchangeably with process model *quality*. Multiple aspects of complexity can be identified and analyzed depending on the purpose of the research and the interpretation of the authors. We select list of measures from existing literature [Car06, CMNR06, SGRM12, Men07] focused on model size and control flow complexity of a given process model, which we denote by *PM*. The measures are widely used, well-defined (i.e., have formal definitions), and theoretical or empirical validations of the measures was conducted [SGRV10].

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The definitions of the selected measures are taken from [Mel12] and are adjusted to fit BPMN-based process models<sup>3</sup> instead of Event-Driven Process Chains (EPCs). The mapping from EPC to BPMN 2.0 is performed based on set of rules introduced in [Tsc06]. Note that we purposefully provide somewhat informal definitions, as the notational overhead of formalizing a practically reasonable subset of BPMN 2.0 would require substantial notational overhead.

**(a) Size Measures.** These measures show how large the model is. The size of the model can be considered as a potential error determinant if the model is produced by a human modeler. These measures serve as a good indicator for the understandability and errors prediction in the model [MMN<sup>+</sup>06], i.e., larger models are more difficult to understand and have a higher error probability than small models [MRv10].

- *TNT* denotes the *total number of tasks* in *PM*;
- *TNE* denotes the *total number of events* in *PM*;
- *TNG* denotes the *total number of gateways* in *PM*;
- *TNP* denotes the *total number of pools* in *PM*;
- *TNL* denotes the *total number of lanes* in *PM*.

**(b) Control-Flow Complexity Measures.** While these size measures are very simple, it is important to consider other aspects of the process, e.g., the control-flow [VCM<sup>+</sup>07]. The complexity of a gateway relates to its type (e.g., exclusive, inclusive, and parallel) and to whether it is a split (i.e., decision point) or a join (e.g., synchronization point). We overtake the gateway complexity

<sup>2</sup>Cf. <https://www.signavio.com/post/improving-process-models-with-modeling-conventions/>, accessed-at 2025-09-09.; specially, we apply *Signavio Best Practices for BPMN 2.0*, as well as *BPMN Method and Style*.

<sup>3</sup><https://www.omg.org/spec/BPMN/2.0>

indicator introduced in [SGGRM12] along with the measures utilized for its calculation to consider different aspects of the control flow.

Assuming that  $G$  is the set of all existing gateways in the process model  $PM$ ,  $S$  denotes the set of split gateways and  $J$  the set of joints ( $G = S \cup J$ ). To address a subset of gateways of a special type (e.g., exclusive, inclusive, and parallel), an additional label is used as an index (e.g.,  $S_{exc}$  denotes the set of splitting exclusive gateways). Each gateway  $g \in G$  has degree  $d(g)$  as a property. The degree  $d(g)$  is the number of incoming and outgoing arcs of a sequence flow of the gateway:  $d(g) = d_{in}(g) + d_{out}(g)$ .

We are aware that according to the BPMN2.0 Specification, there are more gateway types than the three introduced here (e.g., event-based, complex, etc. gateways). However, since our datasets are restricted to include only the three mentioned above (see Table 3.1), and approaches (see Section 3.2) are limited to the basic set of BPMN elements, we restrict the set of supported gateway types to the selected three. If the set of gateway types has to be extended, it also has to be considered in the measures and has to be adjusted correspondingly (e.g., the number of gateway types influences the gateway heterogeneity, i.e., the base of the logarithm).

- Control-Fow Complexity is the sum over all decision points in  $PM$  based on their type and the number of outgoing transitions [Car05]:

$$CFC = \sum_{g \in S} \begin{cases} 1, & \text{if } g \text{ is a splitting parallel gateway in } S_{par} \\ d_{out}(g), & \text{if } g \text{ is a splitting exclusive gateway in } S_{exc} \\ 2^{d_{out}(g)} - 1, & \text{if } g \text{ is a splitting inclusive gateway in } S_{inc} \end{cases}$$

- Gateway Mismatch is the sum of gateway pairs that do not match with each other inside  $PM$  [Men08]:

$$GM = \sum_{l \in \{par, exc, inc\}} \left| \sum_{g \in S_l} d_{out}(g) - \sum_{g \in J_l} d_{in}(g) \right|$$

- Gateway Heterogeneity is the frequency of gateway types used in  $PM$  [Men08]:

$$GH = - \left( \sum_{l \in \{par, exc, inc\}} \frac{|G_l|}{TNG} \cdot \log_3 \frac{|G_l|}{TNG} \right)$$

- Average Gateway Degree is the average degree value over all gateways in  $PM$  [Men08]:

$$AGD = \frac{1}{TNG} \sum_{g \in G} d(g)$$

- Maximum Gateway Degree is the maximum degree value  $d(g)$  over all gateways in  $PM$  [Men08]:

$$MGD = \max_{g \in G} d(g)$$

- Gateway Complexity Indicator is a unified value intended to summarize multiple gateway complexity measures for  $PM$  [SGGRM12]:

$$GCI = 0.176 \times CFC + 0.177 \times GM + 0.159 \times GH + 0.175 \times AGD + 0.180 \times MGD + 0.179 \times TNG$$

[T: What is the motivation of the weights?]

**Process Model Completeness.** Even though the process complexity measures might serve as a possible indicator of process model quality, these measures do not provide an overview of its semantic aspects. When generating a process model from text using genAI, the semantic alignment between the source process description and the generated process model is important. So far, there are no measures that can ensure that the generated process model is semantically correct with the process model (except for human expertise). Thus, to evaluate these aspects, we utilize several similarity measures. We measure the similarity between the generated process model and the ground truth model, and between the generated process model and the source process description, to evaluate whether the generated model covers the same content and to what extent (i.e., process model completeness).

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**(a) Model-to-Model Similarity.** To measure Model-to-Model similarity (M2M) we define 3 different measures based on both node and edge matching similarity, which relies on pairwise comparisons of node labels and attribute as well as edges between these nodes in the model:

- **Element-Based Similarity (EBSM)** is the similarity measure introduced in [VHK<sup>+</sup>24], where the similarity score between two BPMN models is calculated by first categorizing the elements in the models by their type (e.g. tasks, events, gateways, etc.) and then grouping the elements based on additional characteristics relevant for each type of element (labels and subtypes). For each group (subset) of elements, semantically similar elements in the two models are identified using the Dice score [Dic45]. The overall score is then weighted by the arithmetic mean of the length of the elements in the group. The measure considers flow objects, swimlanes, and connecting objects: task labels and types, event labels and types, gateway labels and types, labels of lanes and pools, as well as sequence and message flow edges (i.e., labels of source and target elements).

$$EBSM(M, M') = \frac{\sum_{s \in S} w_s \times sim_s(M_s, M'_s)}{\sum_{s \in S} w_s}, \text{ where}$$

- $M$  is a ground truth process model,
- $M'$  is a generated process model,
- $S$  is a set of possible groups (subsets) of the elements aggregated by type and characteristics (e.g., tasks labels, event type, etc.),
- $w_s$  is the number of the elements in each subset  $s \in S$ :

$$w_s = |M_s| + |M'_s|$$

- $sim_s(M_s, M'_s)$  is the similarity score between two subsets of model elements in particular group  $s \in S$  in models  $M$  and  $M'$  calculated via the Dice score:

$$sim_t(M_s, M'_s) = Dice(M_s, M'_s) = \frac{2 \times |M_s \cap M'_s|}{|M_s| + |M'_s|}$$

- **Label-Based Similarity (LBSM)** is the simplified version of *EBSM* measure, where as the element groups we consider only flow objects and swimlanes: task labels, event types, gateway types, and labels of lanes and pools.
- **Flow-Based Similarity (FBSM)**: similar to *EBSM*, we categorize the elements in the models by type and compare them, identifying semantically similar elements between the two models using the dice score. However, we weight the elements not by their length but rather by giving a 50/50 weighting to the flow elements (sequence flow, message flow) as opposed to non-flow elements (tasks, events, gateways, pools, including lanes) to capture the differences in the control flow of a BPMN model more accurately.

$$FBSM(M, M') = \frac{1}{2} \cdot EBSM(M_f, M'_f) + \frac{1}{2N} \cdot \sum_{nf \in S_{nf}} EBSM(M_{nf}, M'_{nf}), \text{ where}$$

- $S$  is a set of possible groups (subsets) of the elements aggregated by type and characteristics (e.g., tasks labels, event type, etc.),
- $S = S_f \cup S_{nf}$ ,
- $S_f$  is a subset of  $S$  aggregated by type and characteristics where types refer to flow elements,  $f \in S_f$ ,
- $S_{nf}$  is a subset of  $S$  aggregated by type and characteristics where types refer to non-flow elements,  $nf \in S_{nf}$ ,
- $N = |S_{nf}|$  is the number of considered non-flow element groups (i.e., with at least one element present in one of two models)

**(b) Model-to-Text Similarity.** To measure Model-to-Text similarity (M2T) we define different measures serving to show how much content from the process description was used to generate the process model, and whether the process model contained some information, that was not defined in the process description (i.e., hallucinations).

Let  $D$  be a set of sentences in a process description. Let  $M'$  be a generated process model, with  $T'$  is the set of tasks in this model.

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- **Relevance** (i.e., recall) shows how many sentences in the process description  $D$  are relevant for the generated process model  $T'$  (i.e., which sentences in the process description  $d \in D$  are aligned with the tasks  $t' \in T'$  in the generated process model  $M'$ ):

$$Relevance = \frac{D \cap T'}{D}$$

- **Truthfulness** (i.e., precision) shows how many tasks  $T'$  in the generated process model  $M'$  are found in the sentences of the process description  $D$  (i.e., which tasks  $t' \in T'$  in the generated process model  $M'$  are aligned with the sentences in the process description  $d \in D$ ):

$$Truthfulness = \frac{D \cap T'}{T'}$$

- **Ignorance** (i.e., false negative rate) shows how many sentences in the process description  $D$  are ignored during generation of process model  $M'$  (i.e., which sentences in the process description  $d \in D$  are not aligned with the tasks  $t' \in T'$  in the generated process model  $M'$ ):

$$Ignorance = 1 - Relevance = \frac{D \setminus T'}{D}$$

- **Hallucination** (i.e., false discovery rate) shows how many tasks  $T'$  in the generated process model  $M'$  are not similar to the sentences in the process description  $D$  (i.e., which tasks  $t' \in T'$  in the generated process model  $M'$  are not aligned with the sentences in the process description  $d \in D$ ):

$$Hallucination = 1 - Truthfulness = \frac{T' \setminus D}{T'}$$

**Modelling Convention.** Modelling convention is a set of rules defined to ensure the quality and readability of process models, introducing a uniform understanding of modeling styles. By default, the Signavio Process Editor supports two modeling conventions: Signavio Best Practices for BPMN 2.0 and BPMN Method and Style. These modeling conventions go beyond just BPMN 2.0 syntax rules, they also check the model against guidelines derived from analysis of expert opinions, customer experiences and organization-specific modeling guidelines<sup>4</sup> together with a set of rules defined in [Sil11] to create “good BPMN models”, meaning models that are correct, clear, correct, and consistent [Kam20].

Compliance check against these conventions is automatically executed, ensuring their effective usage, ensuring that created models are not only syntactically correct, but also do not suffer from process layout, structure, naming, architecture, and notation issues<sup>5</sup>. The set of layout rules outlines how to properly distance elements, maintain an appropriate diagram size, and correctly define the edge direction of a model. The list of process structure rules defines the semantic and syntactic conventions for process models (e.g., potential deadlocks). Naming deals with process element labels to ensure their consistency. The process architecture convention defines the meta-structure and supporting components of a process model (i.e., how to incorporate open comments, activity descriptions, dictionary links, etc.). Finally, process notation ensures that modeling elements are used in the correct way according to their meaning (e.g., using an AND gateway for parallel execution, and not a XOR one).

After compliance checking against above-mentioned modelling rules the results are split into three groups: **errors, warnings, and hints**. The errors are mandatory to fix, as otherwise the model cannot be considered as a valid one. The warning messages are recommendations, calling your attention to the conditions that might lead to a problem if not resolved. The information messages are more like a hint improving the quality of your model but are not a potential error.

<sup>4</sup><https://www.signavio.com/downloads/short-reads/guide-designing-optimal-business-processes/>

<sup>5</sup><https://www.modeling-guidelines.org/>

### 3.3.3 Performance-related Measures

1. Average Cost per run
2. Average Time per run
3. Other considerations (security, stability, errors)

## 4 Evaluation Results

### 4.1 Basic Capabilities

Table 2: Comparison of Basic Capability Dimensions

| Feature                 | PromoAI    | KLU        | AutoBPMN            | SplitMiner | Trees |
|-------------------------|------------|------------|---------------------|------------|-------|
| BPMN 2.0 Support        |            | ✓          | ✓                   | ✓          | ✓     |
| Stencils Covered, %     |            |            |                     |            |       |
| User Interface          | ✓          | ✓          | ✓                   | ✓          | ✓     |
| Model Redesign via GUI  |            |            |                     |            |       |
| Model Redesign via Chat | ✓          | ✓          | ✓                   |            |       |
| Multiple LLMs           | ✓          | ✓          | ✓                   | ✓          | ✓     |
| Deployment Type         | standalone | standalone | demo                | integrated |       |
| REST API                |            |            | ✓                   |            |       |
| Input Format            |            |            | text                |            |       |
| Output Format           |            |            | custom, image, text |            |       |

### 4.2 Process Model-related Measures

The evaluation is done in two stages. In both stages, we give preference to the GPT models, as they are the widely used LLMs among developers and in industrial applications [SEM25, Sta25].

During the *first stage*, we performed a single run of process model generation with **gpt-4o** (being a widely used powerful general-purpose LLM) to establish a real-world performance baseline.

In the *second stage*, we move towards **gpt-4.1**, as an evolution of the previous model in coding and instruction following and having larger context windows, being oriented towards developer applications [FFF<sup>+</sup>25]. We conducted 10 runs to assess output consistency and variance, allowing a more comprehensive evaluation of LLMs’ reliability and reproducibility, which are essential requirements to integrate LLMs into industrial and research scenarios [WW25].

Comparison between general-purpose gpt-4o and developer-optimized gpt-4.1, on one hand, enables evaluation of how different LLMs handle the process model generation task, and on the other hand, ensures that the approaches being evaluated (and the prompts at their core) are effective, given gpt-4.1 being more ‘literal’ toward instruction interpretation, which means that vague prompts may result in incomplete answers and drop into performance in comparison with the gpt-4o.

#### 4.2.1 Stage One: Single Run

**Number of Generated Models.** First of all, we examine whether all provided process descriptions can be converted into process models by each of the approaches. If any approach results in consistent failures during process model generation, this reflects weaknesses in a particular approach, even if the remaining generated models are of good quality. It can be seen, that in most cases all process descriptions were successfully transformed into process models.

[N: Based on the results, it can be observed that only two approaches show a tendency to fail during model generation. In case of KLU approach these failures can be explained by ... . For the Signavio split approach the reasons are ... .]

@christian: explain the reasons for failures.

However, due to the nondeterministic nature of LLMs, a single run is not sufficient to determine whether these failures are systematic or rather incidental. A more detailed analysis based on multiple runs is required to make reliable conclusions about the approaches.

| Dataset        | Domains | MAD150 | PET35 | SAPSAM69 | RealSet |
|----------------|---------|--------|-------|----------|---------|
| autobpmn       | 1       | 1      | 1     | 1        | 1       |
| klu            | 1       | 0.99   | 1     | 0.97     | 1       |
| promoai        | 1       | 1      | 1     | 1        | 1       |
| signavio       | 1       | 1      | 1     | 1        | 1       |
| signavio_split | 0.92    | 0.98   | 1     | 1        | 1       |

For further evaluation, all data in the tables represent average results calculated only for the models that were successfully generated and converted into BPMN2.0 notation.

**(a) Process Model Complexity: Size Measures.** Looking at the results presented in the Table 3 for high-consistency datasets, we can see that there are no significant differences between the ground truth process models and the generated ones. All approaches tend to create one start and end event.

Table 3: Process Model Complexity: Size Measures. High Consistency Datasets

| Dataset        | Domains      |      |             |      |             | MAD150       |      |             |      |             |
|----------------|--------------|------|-------------|------|-------------|--------------|------|-------------|------|-------------|
| #Elem.         | #T           | #E   | #G          | #P   | #L          | #T           | #E   | #G          | #P   | #L          |
| GT             | 10.28        | 2.00 | 5.36        | 0.00 | 0.00        | 9.79         | 2.00 | 5.79        | 0.00 | 0.00        |
| autobpmn       | 10.64        | 2.00 | 5.76        | 0.00 | 0.00        | 9.91         | 2.00 | 8.12        | 0.00 | 0.00        |
| klu            | 10.84        | 2.00 | 6.00        | 0.00 | 0.00        | 9.63         | 2.00 | 5.81        | 0.00 | 0.00        |
| promoai        | <b>11.20</b> | 2.00 | 5.68        | 0.00 | 0.00        | <b>11.09</b> | 2.00 | 4.84        | 0.00 | 0.00        |
| signavio       | 10.24        | 2.00 | <b>3.68</b> | 1.00 | 1.48        | 9.83         | 2.00 | <b>4.40</b> | 1.00 | 1.47        |
| signavio_split | 9.35         | 2.00 | <b>3.91</b> | 1.04 | <b>1.96</b> | 9.79         | 2.00 | 6.07        | 1.14 | <b>3.03</b> |

The PromoAI approach tends to generate more tasks than other approaches. Since the process descriptions are generated with a template-based approach based on the process model, this can indicate that PromoAI allows hallucinations and that LLMs generate some tasks that are not provided in the process description, or that it has issues with the granularity of the tasks, splitting existing tasks into multiple ones.



Both Signavio approaches tend to generate fewer gateways in comparison to ground truth and other approaches. However, this does not indicate that there is a lower number of decisions present. It can rather indicate that, in some cases, the flow is not merged back after the decision split, since it is not strictly required by the BPMN standard. Academic approaches puts more value on the block-structured models. However, Signavio approaches generate pools and lanes. The Signavio Split approach generates more lanes than pools, in comparison to the Signavio approach.

Table 4: Process Model Complexity: Size Measures. Intermediate Consistency Datasets

| Dataset        | PET35       |             |             |      |             | SAPSAM69     |             |             |      |             |
|----------------|-------------|-------------|-------------|------|-------------|--------------|-------------|-------------|------|-------------|
| #Elem.         | #T          | #E          | #G          | #P   | #L          | #T           | #E          | #G          | #P   | #L          |
| GT             | 10.40       | 2.29        | 4.34        | 1.06 | 2.63        | 9.25         | 3.65        | 3.55        | 1.46 | 2.09        |
| autobpmn       | 8.71        | <b>2.14</b> | 5.20        | 0.00 | 0.00        | 10.78        | <b>2.16</b> | 6.06        | 0.00 | 0.00        |
| klu            | 8.74        | 2.00        | 4.74        | 0.00 | 0.00        | 11.03        | 2.00        | 5.10        | 0.00 | 0.00        |
| promoai        | <b>9.80</b> | 2.00        | 4.69        | 0.00 | 0.00        | <b>13.74</b> | 2.00        | 5.74        | 0.00 | 0.00        |
| signavio       | 7.97        | 2.00        | <b>1.66</b> | 1.00 | 2.11        | 10.71        | 2.00        | <b>3.36</b> | 1.00 | 2.00        |
| signavio_split | 8.71        | 2.00        | <b>2.46</b> | 1.23 | <b>2.57</b> | 11.93        | 2.00        | 4.14        | 1.09 | <b>2.58</b> |

The same behavior is also relevant for the intermediate datasets (see Tab. 4), with the one difference that in the AutoBPMN approach the number of events is slightly increased. Since the implementation of the approach does not allow having more than one start and end event, it indicates the usage of other events (only two options are possible: terminate or escalate), meaning that the complexity of the generated process slightly increases in comparison to the previous datasets.

Table 5: Process Model Complexity: Size Measures. Low Consistency Datasets

| Dataset        | RealSet |             |             |      |             |
|----------------|---------|-------------|-------------|------|-------------|
| #Elem.         | #T      | #E          | #G          | #P   | #L          |
| GT             | 14.52   | 4.30        | 8.22        | 1.78 | 2.04        |
| autobpmn       | 9.26    | <b>2.09</b> | 5.61        | 0.00 | 0.00        |
| klu            | 11.43   | 2.00        | 5.65        | 0.00 | 0.00        |
| promoai        | 9.83    | 2.00        | 5.04        | 0.00 | 0.00        |
| signavio       | 7.74    | 2.00        | <b>1.04</b> | 1.00 | 1.52        |
| signavio_split | 9.35    | 2.00        | <b>2.83</b> | 1.17 | <b>2.57</b> |

Considering the dataset with low consistency (see Tab. 5), it is interesting that none of the approaches generate as many tasks as are provided in the ground truth (up to five tasks). As we know that there is a large discrepancy between the process model and the process description in this case, we can conclude that these tasks are not provided in the process description and were added by the process modeler during the modeling process.

We can sum up that, despite the different nature of the data, the behavior of the LLM remains consistent across these datasets and across various approaches.

**(b) Process Model Complexity: Control Flow.** In the next three tables 6, 7, 8 we present detailed results for the control-flow evaluation. For more details, see Section ?? . Based on the ground truth, we can observe that all models are fairly easy to understand and modify (for more details, see [?] ).

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Substantial differences are observed in only three metrics (CFC, GM, and GH) out of the six. Regarding the CFC index, lower values indicate simpler control-flow and a lower number of execution paths (mental states) occurring in the model. As long as the index remains low, no significant differences are observed between the ground-truth and LLM-generated models. However, once the index increases slightly, differences become tangible, i.e., LLM-generated models tend to become simpler than the ground-truth ones (e.g., see CFC value for MAD150 or RealSet). This may indicate that LLMs ignore, miss or abstract some potential decision points. At the same time, it means that certain decisions were possibly introduced in the ground-truth models without being explicitly specified in the process description.

Table 6: Process Model Complexity: Gateway Complexity. High Consistency

| Datasets<br>Dataset | Domains |      |      |      |      |      | MAD150      |      |      |      |      |      |
|---------------------|---------|------|------|------|------|------|-------------|------|------|------|------|------|
| Measure             | CFC     | GM   | GH   | AGD  | MGD  | GCI  | CFC         | GM   | GH   | AGD  | MGD  | GCI  |
| GT                  | 4.92    | 0.00 | 0.15 | 2.98 | 3.52 | 3.01 | <b>6.65</b> | 0.00 | 0.29 | 2.92 | 3.16 | 3.33 |
| autobpmn            | 4.72    | 0.16 | 0.16 | 3.04 | 3.52 | 2.99 | 3.83        | 0.01 | 0.25 | 2.92 | 3.16 | 2.85 |
| klu                 | 5.44    | 0.36 | 0.15 | 2.89 | 3.00 | 3.17 | 3.28        | 0.01 | 0.17 | 2.90 | 3.14 | 2.74 |
| promoai             | 5.36    | 0.16 | 0.16 | 3.04 | 3.68 | 3.21 | 3.42        | 2.77 | 0.10 | 3.37 | 4.42 | 3.36 |
| signavio            | 3.32    | 0.00 | 0.12 | 2.91 | 2.96 | 2.30 | 2.63        | 0.00 | 0.06 | 2.86 | 3.02 | 2.31 |
| signavio_split      | 3.24    | 0.40 | 0.13 | 2.95 | 3.16 | 2.60 | 4.04        | 0.77 | 0.23 | 2.99 | 3.37 | 3.14 |

GM value, which counts non-matching gateway pairs, remains consistently low and close to the ideal value of zero in LLM-generated models. In contrast, ground-truth models created by process modelers exhibit higher GM values (e.g., GM reaches up to 3.48 for the Realset dataset).

Table 7: Process Model Complexity: Gateway Complexity. Intermediate Consistency Datasets

| Dataset        | PET35 |      |      |      |      |      | SAPSAM69 |             |      |      |      |      |
|----------------|-------|------|------|------|------|------|----------|-------------|------|------|------|------|
| Measure        | CFC   | GM   | GH   | AGD  | MGD  | GCI  | CFC      | GM          | GH   | AGD  | MGD  | GCI  |
| GT             | 4.23  | 0.06 | 0.09 | 2.44 | 2.60 | 2.44 | 3.88     | <b>1.90</b> | 0.21 | 2.88 | 3.04 | 2.74 |
| autobpmn       | 5.17  | 0.34 | 0.05 | 2.52 | 2.69 | 2.81 | 5.28     | 0.25        | 0.13 | 2.80 | 2.99 | 3.02 |
| klu            | 4.60  | 0.00 | 0.16 | 2.57 | 2.71 | 2.62 | 4.43     | 0.03        | 0.15 | 2.83 | 2.96 | 3.24 |
| promoai        | 4.31  | 0.51 | 0.28 | 2.65 | 3.03 | 2.74 | 5.19     | 0.28        | 0.23 | 2.97 | 3.30 | 3.13 |
| signavio       | 1.57  | 0.00 | 0.04 | 2.06 | 2.06 | 1.31 | 3.07     | 0.00        | 0.09 | 2.78 | 2.81 | 0.83 |
| signavio_split | 2.51  | 0.29 | 0.04 | 2.56 | 2.77 | 1.89 | 3.94     | 0.64        | 0.12 | 2.95 | 3.20 | 2.09 |

LLM-generated models are generally characterized by more uniform gateway types, reflected by consistently lower GH values compared to ground-truth models, regardless of the dataset.

The remaining metrics remain relatively constant and do not show significant differences between LLM-generated and ground-truth models.

Not all LLM-based approaches behave identically with respect to gateway complexity metrics. Models generated using the Signavio approach consistently achieve the lowest values across all metrics and datasets. Academic approaches (AutoBPMN, KLU, and PromoAI), in contrast, return models with more complex control flow.

Overall, LLM-based approaches tend to simplify control-flow structures, resulting in lower gateway complexity, which on one hand ensure more stable and correct solutions with higher understandability, but on the other hand carries in the risk of losing decision logic present in proposed process descriptions.

Table 8: Process Model Complexity: Gateway Complexity. Low Consistency Datasets

| Dataset        | RealSet     |             |      |      |      |      |
|----------------|-------------|-------------|------|------|------|------|
| Measure        | CFC         | GM          | GH   | AGD  | MGD  | GCI  |
| GT             | <b>8.91</b> | <b>3.48</b> | 0.56 | 3.11 | 3.52 | 4.92 |
| autobpmn       | 5.39        | 0.17        | 0.12 | 3.22 | 3.35 | 3.12 |
| klu            | 5.30        | 0.00        | 0.26 | 3.34 | 3.70 | 2.81 |
| promoai        | 4.61        | 0.48        | 0.33 | 3.26 | 3.91 | 3.14 |
| signavio       | 1.00        | 0.00        | 0.00 | 1.30 | 1.30 | 2.15 |
| signavio_split | 2.70        | 0.22        | 0.09 | 2.82 | 3.13 | 2.66 |

Low values of GH across all generated models indicate that there are almost no mismatched gateways, meaning that the number of split and merge gateways is roughly equally divided, i.e., the number of both splitting and merging gateways is around the half of all gateways ( $TNG/2$ ). Given this, it can be seen that in all generated models, the total number of gateways (TNG) is very similar to the CFC, which implies that CFC is approximately twice the number of splitting gateways, since the number of splitting gateways is  $TNG/2$ . The fact that CFC is approximately twice the number of splitting gateways suggests that most gateways are exclusive (XOR) splits, while parallel (AND) and inclusive (OR) gateways are used rarely. This is further supported by the consistently low GH values, showing that the generated models rely primarily on a small, uniform set of gateway types.

**(a) Process Model Completeness: M2M Similarity.** Based on the results presented in Tables ?? we can see that LBSM values are higher than EBSM and FBSM across all datasets. The decrease in values between LBSM and EBSM means that the difference in the models is mostly due to the types of the elements and not their labels, meaning that LLMs agree on what the generated models are about, but do not always agree on how those models and the elements within them are declared and connected.

Table 9: Process Model Similarity: M2M. Low and High Consistency

| Dataset        | Domains |      |      | MAD150 |      |      | RealSet |      |      |
|----------------|---------|------|------|--------|------|------|---------|------|------|
| Measure        | LBSM    | EBSM | FBSM | LBSM   | EBSM | FBSM | LBSM    | EBSM | FBSM |
| autobpmn       | 0.96    | 0.92 | 0.91 | 0.78   | 0.58 | 0.67 | 0.45    | 0.29 | 0.36 |
| klu            | 0.96    | 0.89 | 0.87 | 0.85   | 0.58 | 0.61 | 0.46    | 0.30 | 0.32 |
| promoai        | 0.95    | 0.90 | 0.90 | 0.82   | 0.62 | 0.69 | 0.44    | 0.28 | 0.32 |
| signavio       | 0.95    | 0.66 | 0.75 | 0.88   | 0.74 | 0.75 | 0.26    | 0.24 | 0.22 |
| signavio_split | 0.95    | 0.66 | 0.74 | 0.84   | 0.70 | 0.74 | 0.26    | 0.24 | 0.23 |
| range          | 0.01    | 0.26 | 0.17 | 0.10   | 0.16 | 0.14 | 0.20    | 0.06 | 0.14 |

The observation that EBSM and FBSM are very close in value, with only minor differences (typically between 0.01–0.05), implies that neither differences in element types nor deviations in control-flow (the sequence of elements in the process model) dominate the dissimilarity and are instead balanced.

Notably, in previous evaluations, there were no significant differences in LLM behavior across datasets. Here, we can see that as soon as the complexity of the dataset increases, the similarity values decrease. Thus, for the highly consistent dataset, LBSM similarity achieves between 78 and 96%, whereas for the intermediate datasets LBSM reaches only between 54 and 70%, and for the low-consistency dataset this value reaches only between 26 and 46%. The same trend also applies to EBSM and FBSM.

Table 10: Process Model Similarity: M2M. Intermediate Consistency

| Dataset        | PET35 |      |      | SAPSAM69 |      |      |
|----------------|-------|------|------|----------|------|------|
| Measure        | LBSM  | EBSM | FBSM | LBSM     | EBSM | FBSM |
| autobpmn       | 0.67  | 0.65 | 0.66 | 0.70     | 0.63 | 0.62 |
| klu            | 0.70  | 0.64 | 0.63 | 0.70     | 0.65 | 0.63 |
| promoai        | 0.63  | 0.61 | 0.62 | 0.65     | 0.56 | 0.55 |
| signavio       | 0.55  | 0.41 | 0.48 | 0.64     | 0.50 | 0.51 |
| signavio_split | 0.54  | 0.43 | 0.53 | 0.57     | 0.47 | 0.49 |
| range          | 0.16  | 0.24 | 0.18 | 0.13     | 0.18 | 0.15 |

Relatively high performance for the high-consistency dataset indicates that LLMs can successfully extract existing BPMN constructs and arrange them in a correct sequence of elements. However, based on the results from intermediate-consistency datasets, we can see that as soon as the complexity of the process descriptions increases, this extraction becomes more challenging.

For the low-consistency datasets, we can derive that, on the one hand, the performance decreases, but on the other hand, the existing ground truth is not appropriate for evaluation, and other methods have to be used to evaluate this data. This is confirmed by the metrics presented in previous sections, where we can see the large difference between the number of tasks in generated models and the ground truth, as well as the number of branches and decisions that are present in those models.

Complexity metrics, similar to control-flow complexity metrics, also indicate that not all LLM-based approaches behave identically. For the low-consistency datasets, the difference between approaches is insignificant. However, as soon as the complexity of the datasets increases, the range of the values between approaches varies. The range between similarity metrics for the intermediate and low-consistency datasets reaches up to 24%. Here, we can also see that academic approaches (AutoBPMN, KLU, and PromoAI), generate process models with higher similarity to the ground truth.

**(b) Process Model Completeness: M2T Similarity.** Based on the tables ??, it can be seen that the level of trustfulness of the models generated by LLM models is consistently very high, meaning that the tasks in the generated models are indeed aligned with the process models and are not information coming from the LLM. In other words, when following precise instructions, LLMs do not tend to create hallucinations. This is relevant for all approaches regarding the dataset.

Considering relevance, there is a trend that only around 70% of the whole process description (in some cases even lower, see the Domains dataset) was

Table 11: Process Model Similarity: M2T. Low Consistency

| Dataset        | Domains |        |        |       | MAD150 |        |        |       |
|----------------|---------|--------|--------|-------|--------|--------|--------|-------|
| Measure        | Rel.    | Truth. | Ignor. | Hall. | Rel.   | Truth. | Ignor. | Hall. |
| auto           | 0.53    | 1.00   | 0.47   | 0.00  | 0.73   | 0.97   | 0.27   | 0.03  |
| klu            | 0.53    | 1.00   | 0.47   | 0.00  | 0.70   | 0.98   | 0.30   | 0.02  |
| promoai        | 0.53    | 1.00   | 0.47   | 0.00  | 0.74   | 0.99   | 0.26   | 0.01  |
| signavio       | 0.53    | 1.00   | 0.47   | 0.00  | 0.73   | 0.98   | 0.27   | 0.02  |
| signavio_split | 0.54    | 1.00   | 0.46   | 0.00  | 0.73   | 0.98   | 0.27   | 0.02  |
| range          | 0.01    | 0.00   | 0.01   | 0.00  | 0.04   | 0.02   | 0.04   | 0.02  |

considered for the process description. However, since we consider only task labels, it is possible that some parts of the process description contain information about actors, events, gateways, etc. However, the high consistency of the relevance and ignorance across multiple approaches for each particular dataset indicates that the amount of the used information to derive the tasks is the same everywhere.

Table 12: Process Model Similarity: M2T. Intermediate Consistency

| Dataset        | PET35 |        |        |       | SAPSAM69 |        |        |       |
|----------------|-------|--------|--------|-------|----------|--------|--------|-------|
| Measure        | Rel.  | Truth. | Ignor. | Hall. | Rel.     | Truth. | Ignor. | Hall. |
| auto           | 0.76  | 0.99   | 0.24   | 0.01  | 0.72     | 1.00   | 0.28   | 0.00  |
| klu            | 0.76  | 1.00   | 0.24   | 0.00  | 0.72     | 1.00   | 0.28   | 0.00  |
| promoai        | 0.77  | 0.99   | 0.23   | 0.01  | 0.79     | 1.00   | 0.21   | 0.00  |
| signavio       | 0.74  | 1.00   | 0.26   | 0.00  | 0.73     | 1.00   | 0.27   | 0.00  |
| signavio_split | 0.76  | 0.99   | 0.24   | 0.01  | 0.76     | 1.00   | 0.24   | 0.00  |
| range          | 0.04  | 0.01   | 0.04   | 0.01  | 0.07     | 0.00   | 0.07   | 0.00  |

Stable results across all approaches for each dataset indicate that all approaches and the LLM in the background identify the same amount of information as relevant, and the differences in the model similarity are mainly the result of differences with the ground truth and the sequence of the information that was extracted.

Table 13: Process Model Similarity: M2T. High Consistency

| Dataset        | RealSet |        |        |       |
|----------------|---------|--------|--------|-------|
| Measure        | Rel.    | Truth. | Ignor. | Hall. |
| auto           | 0.71    | 0.99   | 0.29   | 0.01  |
| klu            | 0.71    | 0.99   | 0.29   | 0.01  |
| promoai        | 0.67    | 1.00   | 0.33   | 0.00  |
| signavio       | 0.63    | 0.99   | 0.37   | 0.01  |
| signavio_split | 0.63    | 1.00   | 0.37   | 0.00  |
| range          | 0.07    | 0.01   | 0.07   | 0.01  |

**Modelling Convention.** According to the modeling convention, errors are severe violations of the modeling language and therefore must be fixed immediately. Warnings and information messages, on the other hand, serve as recommendations and have a more informative character. Therefore, a high number of errors indicates that an approach is not feasible for creating valid BPMN models, while a higher number of warnings or information messages just indicates the presence of issues that can be addressed and does not necessarily

mean that the approach suffers from serious problems.

Based on the data presented in Tables ??, we can see that the number of errors across all approaches and datasets is extremely low. In three out of the five approaches (AutoBPMN, PromoAI, and Signavio) no errors were identified. The other two approaches (KLU and Signavio Split) have fewer than one error per model. All identified errors are related to the structural aspects of the process models, indicating issues with the syntax of the modeling language.

Table 14: Modelling Convention. Low and High Consistency

| Dataset        | Domains     |              |       | MAD150      |              |       | RealSet |              |       |
|----------------|-------------|--------------|-------|-------------|--------------|-------|---------|--------------|-------|
| Message        | Err-s       | Warn-s       | Info  | Err-s       | Warn-s       | Info  | Err-s   | Warn-s       | Info  |
| autobpmn       | 0.00        | <b>19.80</b> | 11.48 | 0.00        | <b>23.45</b> | 10.46 | 0.00    | <b>15.87</b> | 11.00 |
| klu            | <b>0.08</b> | 3.44         | 11.72 | 0.00        | 4.52         | 10.41 | 0.00    | 4.87         | 12.39 |
| promoai        | 0.00        | 7.92         | 12.04 | 0.00        | 9.97         | 11.86 | 0.00    | 8.13         | 10.61 |
| signavio       | 0.00        | 1.60         | 18.24 | 0.00        | 22.93        | 10.61 | 0.00    | 1.43         | 23.22 |
| signavio_split | <b>0.17</b> | 4.22         | 21.13 | <b>0.35</b> | 8.00         | 23.33 | 0.00    | 4.35         | 21.09 |

Most of the errors (59% of all cases) occur because of incorrect use of sequence flows (e.g., sequence flows crossing process-level boundaries due to the absence of required message flows). These errors appear only in Signavio Split approach and it does not exhibit any other types of errors. Remaining errors either refer to edges and nodes that are not connected to subsequent elements (e.g., an edge without a source or a start event without an outgoing sequence flow, 30% of all cases) or to gateways (e.g., gateways with no outgoing or incoming sequence flows, 11% of all cases).

All approaches suffer from warnings. Warnings appear in 4 out of 5 existing categories (i.e., Structure, Layout, Architecture, and Naming).

83% of all warnings occur in Layout. Here, the most common warnings are edge intersection (edge overlays another edge, 54%), edge direction (edge is going in the wrong direction, 18%), edge folding (folding of the edge is not consistent, 8.5%), and node intersection (connecting nodes of an element overlap other connecting nodes, 2.5%).

Such a high number of warnings in Layout can be explained by the AutoBPMN approach. This approach uses a vertical (and not horizontal) orientation of the BPMN diagram, which confuses the built-in rules of the modeling conventions in Signavio and produces warnings, even though the model is still correct. This is also confirmed by the highest number of warnings in this approach in comparison with the others.

9% of all warnings occur in Architecture. All Architecture warnings refer to distinct diagram names, meaning that each diagram should have a different name within a particular process landscape to avoid confusion. This warning is not relevant for the approaches themselves and depends on the local landscape of the user.

5% of all warnings refer to Naming, where the single problem is a missing labels for XOR gateways. This indicates that selected approaches (rather the logic of the approach itself or the LLM in a background) often ignore labels for gateways, which can reduce the readability of the model, since XOR gateways

should always be named consistently. Two approaches (KLU and Signavio), however, perform better than the others, indicating that these approaches consistently label XOR gateways.

The remaining 3% of all warnings occur in Structure. The common errors here are inconsistent usage of start and end events, meaningless gateways (a gateway must always have multiple incoming or outgoing flows, otherwise, the gateway is not necessary), and split and join behavior on a single gateway (no separation between splitting and merging gateways).

Table 15: Modelling Convention. Intermediate Consistency

| Dataset        | PET35       |              |       | SAPSAM69    |              |       |
|----------------|-------------|--------------|-------|-------------|--------------|-------|
| Message        | Err-s       | Warn-s       | Info  | Err-s       | Warn-s       | Info  |
| autobpmn       | 0.00        | <b>17.86</b> | 10.91 | 0.00        | <b>19.86</b> | 13.20 |
| klu            | 0.00        | 3.34         | 9.23  | <b>0.61</b> | 5.39         | 11.78 |
| promoai        | 0.00        | 7.34         | 10.40 | 0.00        | 8.57         | 14.62 |
| signavio       | 0.00        | 2.63         | 22.74 | 0.00        | 2.71         | 26.57 |
| signavio_split | <b>0.11</b> | 3.86         | 20.60 | 0.00        | 7.36         | 25.46 |

All approaches produce information messages. Here, 3 out of 5 categories (Notation, Layout, and Structure) are affected. 95% of all information messages refer to the Notation category. 77% of these messages indicate that mandatory attributes are not set (i.e., each activity requires not only a label but also a description/documentation). 18% of messages indicate that many model elements do not comply with Signavio’s built-in glossary.

5% of the messages inform that the size of the diagram is too large, and the remaining 0.07% pertain to Structure, where it is indicated that there are too many Expanded Pools in a model. The built-in rule is to use only one Expanded Pool, since each Pool represents its own process in BPMN.

It can be seen, that Information Messages are rather related to the internal Signavio rules and requirements, then to the modelling approaches themselves. While they are important for model standardization, and readability within one particular conventions/organization, they must not be interpreted as a measure of the quality or correctness of a modeling approach.

**Summary.** In general, we can see that, regarding process model complexity, both Signavio approaches generate simpler process models with fewer elements and a simplified control flow. This means that some parts of the process description are ignored and not considered in comparison to other approaches (see Ignorance in Tab. ??). This subsequently influence the achieved similarity score between the generated models and the ground truth models, where Signavio approaches achieve lower similarity scores for datasets with higher inconsistency (see Tab. ??). On the other hand, this simplicity potentially affects the results of modelling convention check, where the Signavio approach results in no errors and the lowest number of warnings in comparison to other approaches. However, this is not the case for the Signavio Split approach as it is one of two approaches in which errors were detected. Academic approaches tend to generate models with higher complexity and are also able to produce error free process models.

All approaches produce no hallucinations (see Hallucination in ??) and

achieve good results for model similarity for datasets with high and intermediate consistency, meaning that all approaches are able to interpret the provided process descriptions and extract the necessary information. The level of existing errors in modelling conventions is extremely low and is mostly the result of high errors in a single model rather than a consistent pattern, meaning that the extracted information can generally be arranged into an acceptable process model.

#### Performance Measures

tbd

#### 4.2.2 Stage Two: Multiple Runs

**Number of Generated Models.** Similar to the single-run evaluation, the evaluation is started by examining how many process descriptions were successfully converted into process models by each approach. Presented results is the average value across ten runs.

There are two approaches (AutoBPMN and Signavio) that are consistently reliable and sufficient in generating models for each process description across multiple runs. The remaining three approaches show a tendency to fail in some cases. Interestingly, in the case of the Signavio Split approach, these failures appear to be random in nature and occur only in several runs, whereas KLU and PromoAI show systematic failures in every run, with varying numbers of failures.

| Dataset        | Domains     | MAD150      | PET35       | SAPSAM69    | RealSet     |
|----------------|-------------|-------------|-------------|-------------|-------------|
| autobpmn       | 1.00        | 1.00        | 1.00        | 1.00        | 1.00        |
| klu            | <b>0.92</b> | <b>0.91</b> | <b>0.86</b> | <b>0.89</b> | <b>0.91</b> |
| promoai        | 0.94        | 0.95        | 0.92        | 0.96        | 0.97        |
| signavio       | 1.00        | 1.00        | 0.99        | 1.00        | 1.00        |
| signavio_split | <b>0.92</b> | 0.97        | 0.99        | 1.00        | 0.97        |

[N: Based on the results, the failures can be explained by ... .]

@christian: explain the reasons for failures.

For further evaluation, all data in the tables represent the average of the average results across all ten runs calculated only for the models that were successfully generated and converted into BPMN2.0 notation.

**Repeatability and Reproducibility.** Before moving to the next steps in the evaluation, given that multiple runs were conducted, it is necessary to ensure that the results across multiple runs are consistent. Otherwise, if there is a high volatility in the evaluation results between the runs, the results cannot be averaged and interpreted meaningfully.

To estimate how consistent are results across multiple runs and multiple approaches we adopt concepts of Semantic Repeatability and Semantic Reproducibility from [SRH<sup>+</sup>25].

*Semantic Repeatability* shows how consistent is the meaning of outputs across repeated runs under identical conditions (e.g., LLM, parameters, and prompts). *Semantic Reproducibility* on the other hand demonstrates how consistent is the



meaning of outputs across repeated runs with the same LLM and parameters, but with multiple prompts. In other words, semantic repeatability take in consideration, how similar is the generated output from run to run in the frame of one approach, whereas the semantic reproducibility examines, how similar the generated outputs are across multiple approaches.

More specifically, in the context of process modelling, we define the Semantic Repeatability Score as the average pairwise similarity value across all generated process models, with higher values indicating more consistent semantics of generated process models across repeated runs within one particular approach.

$$Rpt = \frac{2}{R(R-1)} \sum_{1 \leq r < s \leq R} LBSM(m'_r, m'_s), \quad 0 \leq Rpt \leq 1,$$

where  $R$  denotes the number of runs,  $r$  and  $s$  are the indexes of distinct runs,  $m'_r, m'_s$  are generated process models in multiple runs, and  $LBSM$  is a similarity function to measure the similarity between two process models (see Sect. ?? for more details).

Semantic Reproducibility Score is defined as the average pairwise consistency between approaches, computed as one minus the absolute difference between their average model-ground-truth similarity scores across multiple runs. Higher values indicate greater reproducibility of process model generation quality across different approaches employing the same LLM.

$$Rpd = \frac{2}{A(A-1)} \sum_{1 \leq a < b \leq A} \left(1 - |\overline{LBSM}_a - \overline{LBSM}_b|\right), \quad 0 \leq Rpd \leq 1,$$

where where  $A$  denotes the number of approaches,  $a$  and  $b$  are the indexes of distinct approaches, and  $\overline{LBSM}_x$  is an average process model similarity function comparing a generated model  $m'_x$  with the ground truth model  $m$  for each two particular approaches  $a$  and  $b$  over all runs  $R$ .

$$\overline{LBSM}_x = \frac{1}{R} \sum_{r=1}^R LBSM(m, m'_{x,r}), \quad x \in \{a, b\}$$

maybe move part with the formulas to 3.3 and here discuss only results

Since semantic reproducibility relies on multiple outputs from each approach, high semantic repeatability is a prerequisite for meaningful reproducibility. Without consistent outputs across runs, high or low reproducibility between approaches could be a result of variability.

Considering Semantic Repeatability (see Tab. 4.2.2), generally, all approaches demonstrate high values across ten runs, despite nondeterministic nature of the LLMs. The average values across all approaches range between 0.71 to 0.86. There is a clear difference in repeatability across datasets, where the datasets with the high consistency shows higher repeatability, and by lower consistency repeatability drops.

Interestingly, all three academic approaches (AutoBPMN, KLU, and PromoAI) show a consistent decline in repeatability from more to less consistent

| Dataset        | Domains     | MAD150      | PET35       | SAPSAM69    | RealSet     |
|----------------|-------------|-------------|-------------|-------------|-------------|
| autobpmn       | <b>0.94</b> | <b>0.96</b> | <b>0.86</b> | <b>0.89</b> | <b>0.80</b> |
| klu            | 0.89        | 0.88        | 0.75        | 0.81        | 0.70        |
| promoai        | 0.89        | 0.83        | 0.75        | 0.79        | 0.68        |
| signavio       | 0.85        | 0.86        | 0.79        | 0.87        | 0.63        |
| signavio_split | 0.75        | 0.79        | 0.85        | 0.86        | 0.73        |
| average        | 0.86        | 0.86        | 0.80        | 0.84        | 0.71        |

datasets. In contrast, both Signavio approaches exhibit high variability in behavior. Signavio performs consistently well on both high and intermediate consistency datasets but shows the lowest repeatability score on the RealSet (0.63). Signavio Split demonstrates lower repeatability on simpler datasets compared to the other approaches, but achieves good results on both intermediate and low consistency datasets. Such behavior may indicate that this approach uses some unique preprocessing or extraction methods that handle easier, more natural process descriptions well, but encounter difficulties with more artificial texts.

Overall, AutoBPMN demonstrates the highest repeatability values across all datasets, with values ranging from 0.80 to 0.96.

Regarding Reproducibility, it can be seen that scores across all pairs of approaches and datasets are high (mostly above 0.90), indicating strong consistency in model similarity across different approaches using the same LLM. High reproducibility scores for high consistency datasets, and slightly lower reproducibility on intermediate and low consistency datasets, at the same time, suggest that there are some differences in approach strategies, but they are first noticeable in complicated cases.

Interestingly, the reproducibility scores are high even for the inconsistent datasets (where similarity between ground truth and generated process models is low), meaning that the generated models are very similar to each other, even though they are not similar to the ground truth.

| Dataset                   | Domains | MAD150 | PET35 | SAPSAM69 | RealSet |
|---------------------------|---------|--------|-------|----------|---------|
| signavio_split × autobpmn | 0.95    | 0.99   | 0.92  | 0.92     | 0.81    |
| signavio_split × promoai  | 0.99    | 0.97   | 0.94  | 0.94     | 0.80    |
| signavio_split × signavio | 1.00    | 0.93   | 0.97  | 0.93     | 0.97    |
| signavio_split × klu      | 0.97    | 0.99   | 0.93  | 0.92     | 0.98    |
| autobpmn × promoai        | 0.94    | 0.97   | 0.98  | 0.98     | 0.99    |
| autobpmn × signavio       | 0.94    | 0.92   | 0.94  | 0.99     | 0.84    |
| autobpmn × klu            | 0.92    | 1.00   | 0.85  | 1.00     | 0.83    |
| promoai × signavio        | 0.99    | 0.90   | 0.96  | 0.99     | 0.83    |
| promoai × klu             | 0.98    | 0.97   | 0.87  | 0.98     | 0.82    |
| signavio × klu            | 0.98    | 0.92   | 0.90  | 0.99     | 0.99    |
| average                   | 0.97    | 0.96   | 0.93  | 0.97     | 0.89    |

### 4.2.3 Model Measures

**(a) Process Model Complexity: Size Measures.** Generally speaking, the broader tendency remains the same: in the case of high and intermediate consistency datasets, selected approaches generate either the same or a slightly greater/smaller amount of modelling elements. At the same time, in the case of low consistency datasets, the number of generated elements is notably lower

than in the provided ground truth process models.

Table 16: Process Model Complexity: Size Measures. High Consistency

| Datasets<br>Dataset | Domains |      |             |      |      | MAD150 |             |      |      |      |
|---------------------|---------|------|-------------|------|------|--------|-------------|------|------|------|
| #Elem.              | #T      | #E   | #G          | #P   | #L   | #T     | #E          | #G   | #P   | #L   |
| GT                  | 10.28   | 2.00 | 5.36        | 0.00 | 0.00 | 9.79   | 2.00        | 5.79 | 0.00 | 0.00 |
| autobpmn            | 10.39   | 2    | 5.57        | 0    | 0    | 11.19  | 2           | 4.75 | 0    | 0    |
| klu                 | 9.59    | 2    | 5.02        | 1    | 1.12 | 9.39   | <b>2.61</b> | 5.54 | 1.12 | 1.55 |
| promoai             | 10.44   | 2    | 5.09        | 0    | 0    | 10.89  | 2           | 6.73 | 0    | 0    |
| signavio            | 11      | 2    | 5.99        | 1    | 1.13 | 9.86   | 2           | 5.71 | 1    | 2.28 |
| signavio_split      | 9.35    | 2    | <b>3.93</b> | 1.05 | 1.87 | 9.77   | 2           | 6.07 | 1.23 | 3.25 |

Across approaches, the main difference compared to the single run is that the KLU approach was changed by extending the set of supported BPMN elements. That can be seen in the number of generated events, pools, and lanes (see columns #E, #P, and #L in Tab. 18, 17, 16). It is notable, that the number of pools and lanes across the three approaches (KLU, Signavio, and Signavio Split) is similar, meaning that the approaches indicate the same number of actors involved in the processes, but arrange them in a slightly different manner.

Table 17: Process Model Complexity: Size Measures. Intermediate Consistency

| Datasets<br>Dataset | PET35 |          |            |      |      | SAPSAM69 |             |             |      |      |
|---------------------|-------|----------|------------|------|------|----------|-------------|-------------|------|------|
| #Elem.              | #T    | #E       | #G         | #P   | #L   | #T       | #E          | #G          | #P   | #L   |
| GT                  | 10.40 | 2.29     | 4.34       | 1.06 | 2.63 | 9.25     | 3.65        | 3.55        | 1.46 | 2.09 |
| autobpmn            | 10.56 | 2        | 5.11       | 0    | 0    | 13.02    | 2           | 5.28        | 0    | 0    |
| klu                 | 7.78  | <b>8</b> | 3.73       | 2.08 | 3.05 | 10.65    | <b>6.43</b> | 5.01        | 1.48 | 2.44 |
| promoai             | 10.92 | 2        | 4.65       | 0    | 0    | 14.32    | 2           | 4.68        | 0    | 0    |
| signavio            | 10.88 | 2        | 4.4        | 1    | 2.87 | 14.11    | 2           | 5.13        | 1    | 2.57 |
| signavio_split      | 8.88  | 2        | <b>2.5</b> | 1.22 | 2.64 | 11.92    | 2           | <b>4.15</b> | 1.1  | 2.63 |

Signavio Split has the same pattern of simplifying existing process logic flow by generating fewer gateways in the control flow than other approaches, regardless of the dataset.

Table 18: Process Model Complexity: Size Measures. Low Consistency Datasets

| Dataset        | RealSet |              |            |      |      |
|----------------|---------|--------------|------------|------|------|
| #Elem.         | #T      | #E           | #G         | #P   | #L   |
| GT             | 14.52   | 4.30         | 8.22       | 1.78 | 2.04 |
| autobpmn       | 11.98   | 2            | 6.39       | 0    | 0    |
| klu            | 10.36   | <b>11.32</b> | 6.78       | 2.37 | 3.25 |
| promoai        | 12.91   | 2            | 8.38       | 0    | 0    |
| signavio       | 13.29   | 2            | 5.29       | 1    | 3.56 |
| signavio_split | 9.35    | 2            | <b>2.8</b> | 1.37 | 2.52 |

Update cpee bpmn transformation and rerun evaluation for gateway complexity and model completeness

- (b) Process Model Complexity: Control Flow.
- (a) Process Model Completeness: M2M Similarity.
- (b) Process Model Completeness: M2T Similarity.

Modelling Convention.

run again

### 4.3 Summary

## 5 Limitations

Limitations of all approaches (Limitations of using LLMs). Realistically -i real-world data looks different.

Limitations of the measures (subjectivity of metrics of evaluating).

Limitations per approach that we could test & inspect the source code **PromoAI**: - Security aspect of executing code. - Taking quite a long time to execute code (given the retry-approach of non-executable code)

**KLU Approach:**

**SplitMiner Approach:** The problems are of combinatoric nature (as the LLM needs to create a large number (add exact formula) for a parallel gateway with 5 potential paths of just one activity).

**Annotated Process tree approach:**

**BAML Simplified:**

*Remarks:* - created non-referenced XML elements (empty tag) which we corrected for - creates just once-encoded controlflow (had to be added so that we correctly represent the control-flow) - we also added BPMN layouter as postprocessing - escaped wrongly escaped strings and special characters such as &

## 6 Related Work

In the following, relevant literature is presented. First, an overview over *classical NLP approaches* is given. Those methods utilize symbolic NLP methods, such as named entity recognition (NER) or relationship extraction. While these methods perform well, the recent advances of research on LLMs has created new opportunities for our specific text-to-process task. These approaches are presented as *LLM-based approaches*.

needs  
rewrite,  
since this  
is a copy of  
other work

### 6.1 Traditional NLP Approaches

Prior to the emergence of LLMs, text-to-process was widely approached by combining domain knowledge with classical NLP approaches such as relationship or entity extraction. [?] break the task down into a sentence-level analysis of the input text, using Stanford Parser [?] and WordNet [?]. Furthermore, a heuristic is used to generate the process diagram on the text level from the extracted relationships and elements. Friedrich et al. [FMP11] approaches this problem by splitting sentences in the provided text into individual components such as objects and verbs to identify activities and actors. As an example, a pre-defined

list of words that indicate choices in a process models such as "if" were used to construct a process model. Additionally, they present a test set of 47 text-diagram pairs. [?] introduce a method and graphical tool for process elicitation and modeling, combining Group Storytelling with Text Mining and NLP to enhance knowledge externalization and improve process diagram quality.

## 6.2 LLM-based approaches

Recent advancements in the area of generative AI [?] (mostly around Large Language Models (LLMs)), allow for more generally applicable and generalizable solutions that can cope with lower quality text. The application of LLMs to the text-to-process task has sparked an exciting new line of research. Ranging from simple experimentation [?, ?] to approaches with more elaborate prompt engineering, such as [?]. Especially, in [?], the authors outline that prompt engineering can be an effective method to enable LLMs to generate valid BPMN diagrams. In addition to that, [?] show that LLMs can perform a wider variety of tasks in BPM. [?] provide an overview of different approaches to solving the text-to-process task with LLMs. [C: @nataliia - in the Bellan2020SurveyT2P paper, there are "quantitative comparisons" of also approaches we did not re-implement. but these are measures we also broadly cover right?]

In this spirit, [?] utilize prompt engineering techniques to generate an intermediate representation of a BPMN diagram, using the Partially Ordered Workflow Language (POWL). Similar to Block-Structured-Models, POWL provides soundness guarantees for the generated diagrams. However, its structure is not fully linear and thus harder to generate for an LLM. Whereas [?] rephrase the problem as a conversational task. Instead of trying to generate a final BPMN diagram from a single textual description, their method defines a chat interface, which allows the user to gradually refine the process diagram. This approach is especially sensible due to the subjective nature of the text-to-process task. A single description can lead to possible and valid BPMN diagrams, and thus multiple refinement steps might be required to allow the user to exactly specify the structure of the desired BPMN diagram.

The approach of [NEAA<sup>+</sup>25] leverages a combination of LLM-based steps in combination with deterministic algorithms (e.g. BPMN construction, loop filtering). Their approach first "decomposes" the user input and extracts activities and other events. Then, the relations of the extracted activities are resolved and further correction as part of a "self-healing" step.

## 6.3 Evaluation of LLM-based approaches

[T: I would put this into the background section and start here from the perspective of genAI. The intro is supposed to be compelling, fully covering the SOTA and history of NLP-based process model generation can go to the background.]

continue  
Christian  
- maybe  
introduce  
the same  
differentia-  
tion as Bel-  
lan2020SurveyT2P

## 7 Conclusion

All currently investigated approaches to generate BPMN diagrams from textual input leverage LLMs only by applying "context engineering" (i.e. defining a suitable "intermediate representation", performing retries/more elaborate prompting techniques/flows), but do not leverage fine-tuning for this specific task.

Shortcomings from a commercial PoV:

1. Processes are rarely generated "net new" but modified, updated or merged. Therefore, ex-ante checks for similar or related models prior to generation of new models is required. The user could then select a similar model, decide to create a new model or consider the existing model as context.
2. Use of specified/unspecified modeling conventions (e.g. enforcing rules for gateways or naming conventions, rules to link IT-System to entities in the system)
3. Customer specific language and abbreviations
4. Fine-tuning commercially difficult: While fine-tuning models for the text to process task constitutes an interesting line of research to pursue, we decided against pursuing this from industry perspective due to the cost of fine-tuning with constantly evolving foundation models, the need of provisioning AI models in all regions the service is available as well as the difficulty on fine-tuning on non-public data/customer data.

Research beyond Text2Process: The generation of processes from a textual description that contains all relevant details needed only constitutes a small part of the use case of extracting process-relevant data from existing documents/-knowledge bases.

The internal survey conducted with SAP Signavio customers 2023 mentioned earlier with 35 participants revealed that 50% of customers have more than 500 documents in the form of mostly word documents (26%), pdf documents (28%) or Powerpoint Presentations (24%) which incurs high import effort. However, unlike for the Text2Model creation, there are additional challenges ranked more challenging than the pure import of content (only marked as 21% when asked for "most challenging task" compared to 29% who replied "creating relations to existing process relevant content" or "enriching existing content (e.g. adding descriptions) with 24%.

sp Challenges that require further research

1. Multi-modal documents
2. Incomplete, contradicting information about processes in pot. different detail levels
3. Not only documents (but expert opinions, previous models, goals,... need to be considered) while modeling

4. Conversational process modelling as a preferable interaction pattern instead of one-off generations

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